

A Mathematical Theory of Market World Models: Hierarchical Mean-Field Dynamics, Dual Stochastic Decomposition, and Financial Event Operator Algebras

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July 2026 (*Preprint — submitted to arXiv:q-fin.MF*)

Abstract

We develop a rigorous mathematical theory for *market world models*— probabilistic dynamical systems that learn and simulate the full evolutionary law of financial markets rather than extracting statistical correlations through factor models. Our framework rests on four interlocking contributions.

(i) Dual Stochastic Decomposition. We prove that market noise admits a canonical Lévy–Itô decomposition into two orthogonal components: a *physical noise* driven by a Brownian motion (capturing fundamental information uncertainty) and a *behavioral noise* driven by a pure-jump Lévy process (capturing sentiment and herding effects). We derive a Cramér–Rao-type lower bound showing that prediction variance is bounded below by a sum of two irreducible terms arising from each component—a market-structure analogue of the Heisenberg uncertainty principle.

(ii) Financial Event Operator Algebra. Every corporate or macroeconomic event (IPO, merger, interest rate shock, etc.) is modeled as an affine operator on the market state space. We classify all such operators by their algebraic mode and prove that the full event algebra carries the structure of a *topological groupoid*—a generalisation of a semigroup required because mergers and spin-offs alter the dimension of the state space.

(iii) Hierarchical Mean-Field Theory. Real financial markets exhibit simultaneous mean-field game (MFG) and mean-field control (MFC) structures at different scales: nations interact as MFG agents while firms within each nation face a mixed MFG/MFC problem regulated by a central authority. We formulate a

two-level McKean–Vlasov system and prove existence and uniqueness of a *hierarchical Nash equilibrium* under Lipschitz, boundedness, and Lasry–Lions monotonicity conditions.

(iv) Stochastic Stability. Using the stochastic Lyapunov method for systems driven by the combined Brownian-Lévy noise, we prove that the market world model admits a unique invariant measure and establish sharp exponential convergence rates. Market bubbles and crises correspond precisely to local violations of the Lyapunov condition.

Taken together, these results constitute, to the best of our knowledge, the *first* systematic mathematical theory for market world models: a self-contained, axiomatically grounded framework that connects the world-model paradigm of machine learning to the rigorous traditions of stochastic analysis and mean-field game theory in quantitative finance.

Keywords: world model, mean-field game, stochastic differential equation, Lévy process, groupoid, Lyapunov stability, market microstructure, prediction lower bound.

MSC 2020: 91G99, 60G51, 49N80, 22A22, 60H10.

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1 Introduction

1.1 Motivation

The dominant paradigm in quantitative finance rests on two pillars: factor models, which represent asset returns as linear combinations of explanatory signals, and strategy models, which exploit persistent anomalies for profit (Fama, 1970; Ross, 1976). Despite their practical success, these approaches share a fundamental limitation: they learn *correlations* in historical data without learning the underlying *causal dynamics* of financial markets.

A market *world model* takes a different approach. Inspired by the world-model framework of Ha and Schmidhuber (2018) in reinforcement learning—where an agent maintains

an internal model of its environment’s temporal dynamics—a market world model aims to learn the full evolutionary law governing market states, enabling counterfactual reasoning and scenario propagation unavailable to correlation-based methods.

Despite growing empirical interest (AWMLG, 2026), *no systematic mathematical theory* has been established for market world models. Existing work remains either purely algorithmic (lacking theoretical guarantees) or limited to single-scale models (ignoring the hierarchical multi-agent structure fundamental to financial markets). This paper provides the missing foundation.

1.2 Contributions

Our four main contributions, each embodied in one or more formal theorems, are:

- (i) **Dual Stochastic Decomposition** (Section 4). Theorem 4.2 establishes the canonical Lévy–Itô decomposition of market noise into physical and behavioral components. Theorem 4.4 provides a prediction lower bound—an information-theoretic impossibility result showing that no estimator can eliminate both components of forecast uncertainty.
- (ii) **Financial Event Operator Algebra** (Section 5). Definition 5.1 introduces affine event operators. Theorem 5.5 proves that the full algebra of financial events constitutes a topological groupoid.
- (iii) **Hierarchical Mean-Field Equilibrium** (Section 7). Theorem 7.4 is the main result of the paper: existence and uniqueness of a hierarchical Nash equilibrium for the two-level McKean–Vlasov system with a Lasry–Lions monotonicity structure.
- (iv) **Stochastic Lyapunov Stability** (Section 8). Theorem 8.2 establishes existence of a unique invariant measure and exponential convergence; Proposition 8.5 identifies the analytic signature of market bubbles and crashes.

1.3 Related Work

Mean-field games in finance. The MFG framework of Lasry and Lions (2007) and Huang et al. (2006) has been applied to price impact (Carmona and Lacker, 2015), systemic risk (Carmona and Mouzouni, 2023), and optimal liquidation (Carmona and Delarue, 2018). Multi-level MFG has been studied in the context of inter-bank lending (Carmona and Sun, 2018) but without the hierarchical coupling structure we introduce here.

Originality statement. To the best of our knowledge, this paper is the *first* to: (a) formally define a market world model as a probability-measure-valued map on an

infinite-dimensional state space; (b) prove a Cramér–Rao-type lower bound that identifies two structurally distinct and irreducible components of market prediction uncertainty; (c) model the full algebra of financial events as a topological groupoid, capturing the dimension-changing nature of mergers and spin-offs; (d) establish existence and uniqueness of a hierarchical Nash equilibrium in a two-level McKean–Vlasov system that simultaneously exhibits MFG and MFC structures at different scales. Prior work on world models in finance is either empirical without theory (AWMLG 2026) or models only single-scale dynamics (Carmona and Delarue 2018).

Jump processes in finance. Lévy-driven asset prices are classical (Cont and Tankov, 2004; Kou, 2002). The decomposition of price innovations into a diffusion and a jump component has been studied for estimation purposes (Ait-Sahalia and Todorov, 2009), but not as a *structural* duality between physical and behavioral uncertainty.

World models and reinforcement learning. Since Ha and Schmidhuber (2018), world models have been central to model-based RL (Hafner et al., 2020, 2023). Their application to finance is nascent: AWMLG (2026) show that a generative world model improves portfolio optimization; JPMorgan (2019) apply GANs to order-book simulation. No prior work provides a unified mathematical theory.

Groupoids in physics and mathematics. Groupoids generalize groups by allowing morphisms between different objects. Their use in noncommutative geometry (Connes, 1994) and geometric mechanics (Weinstein, 1996) is well-established; their appearance in financial modeling is novel.

2 Probabilistic Foundations

2.1 Probability Space and Filtration

Throughout the paper, we fix a complete probability space $(\Omega, \mathcal{F}, \mathbb{P})$ equipped with a right-continuous, complete filtration $\mathbb{F} = (\mathcal{F}_t)_{t \in [0, T]}$ for a fixed horizon $T \in (0, \infty)$.

We work with two independent sources of randomness:

- A standard $\mathbb{R}^m$ -valued Brownian motion $W = (W_t)_{t \geq 0}$, and
- A Poisson random measure N^η on $[0, T] \times \mathbb{R}^k$ with (non-zero) σ -finite intensity measure $\lambda \otimes \nu^\eta$, where λ denotes Lebesgue measure. The compensated measure is $\tilde{N}^\eta(dt, dz) := N^\eta(dt, dz) - \nu^\eta(dz) dt$.

Both W and N^η are adapted to $\mathbb{F}$ and mutually independent.

2.2 Itô Stochastic Integral: The Correct Formulation

A critical structural observation distinguishes our framework from ordinary differential equation (ODE) based approaches:

Proposition 2.1 (Non-differentiability of Brownian paths). *With probability one, $t \mapsto W_t$ is nowhere differentiable. Consequently, the expression $\sigma_t \cdot (dW_t/dt)$ is meaningless point-wise, and the shorthand notation $dS_t = \mu_t dt + \sigma_t dW_t$ represents an integral equation, not a differential equation.*

Proof. For any t , the increment $h^{-1}(W_{t+h} - W_t) \sim \mathcal{N}(0, h^{-1})$. As $h \rightarrow 0$, the variance $h^{-1} \rightarrow \infty$, so no L^2 limit exists. By the Paley–Wiener–Zygmund theorem, the set of ω for which $W(\omega)$ is differentiable at any single point has measure zero; a countable union argument extends this to all of $[0, T]$. $\square$

The precise meaning of the market evolution equation is therefore:

Definition 2.2 (Itô integral form). The market state process satisfies the *stochastic integral equation*:

$$\mathbf{S}_t = \mathbf{S}_0 + \int_0^t \mu(\mathbf{S}_u, u) du + \int_0^t \sigma_\tau(\mathbf{S}_u) dW_u + \int_0^t \int_{\mathbb{R}^k} z \tilde{N}^\eta(du, dz) + \sum_{\tau_w \leq t} (T_w - I) \mathbf{S}_{\tau_w^-}, \quad (1)$$

where the second and third Itô integrals are defined as $L^2(\Omega)$ limits of simple predictable approximants, and the last sum runs over all financial event times $\tau_w \in [0, t]$.

The following isometry is fundamental:

Proposition 2.3 (Itô isometry). *For any $\mathbb{F}$ -predictable process σ with $\mathbb{E}[\int_0^T \|\sigma_t\|^2 dt] < \infty$,*

$$\mathbb{E} \left[\left\| \int_0^t \sigma_u dW_u \right\|^2 \right] = \mathbb{E} \left[\int_0^t \|\sigma_u\|^2 du \right].$$

The adaptedness requirement $\sigma_u \in \mathcal{F}_u$ encodes the economic constraint of no look-ahead: trading signals may not depend on future information.

2.3 Lévy Processes and the Lévy–Khintchine Formula

Definition 2.4 (Lévy process). A process $L = (L_t)_{t \geq 0}$ is a *Lévy process* if (i) $L_0 = 0$ a.s., (ii) L has independent, stationary increments, and (iii) L is continuous in probability.

Theorem 2.5 (Lévy–Khintchine, Sato 1999). *The characteristic function of L_t has the form $\mathbb{E}[e^{i\xi^\top L_t}] = e^{-t\Psi(\xi)}$, where the Lévy exponent is:*

$$\Psi(\xi) = -ib^\top \xi + \frac{1}{2} \xi^\top C \xi + \int_{\mathbb{R}^k} \left(1 - e^{i\xi^\top z} + i\xi^\top z \mathbb{1}_{|z| \leq 1} \right) \nu(dz),$$

with drift $b \in \mathbb{R}^k$, diffusion matrix $C \in \mathbb{R}^{k \times k}$ positive semi-definite, and Lévy measure ν satisfying $\int_{\mathbb{R}^k} (1 \wedge |z|^2) \nu(dz) < \infty$. The triple (b, C, ν) is the Lévy characteristic and determines L in law.

2.4 Market State Space

Definition 2.6 (Single-asset state vector). The state of a single asset at time t is the vector $s_t = (p_t, v_t, \ell_t, \kappa_t, \iota_t)^\top \in \mathbb{R}^5$, where:

Coord.	Symbol	Economic meaning	Domain
1	p_t	Log-price	$\mathbb{R}$
2	v_t	Log-trading volume	$\mathbb{R}_+$
3	ℓ_t	Leverage ratio	$[0, \infty)$
4	κ_t	Outstanding shares (log)	$\mathbb{R}_+$
5	ι_t	Public information disclosure level	$[0, 1]$

For n assets, the joint state is $\mathbf{S}_t \in \mathbb{R}^{5n}$. In the continuum limit ($n \rightarrow \infty$) relevant for theoretical analysis, we work in the separable Hilbert space $\mathcal{H} = L^2([0, 1] \times [0, T], \mathbb{R}^5)$ with inner product $\langle f, g \rangle = \int_0^1 \int_0^T f(u, t) \cdot g(u, t) dt du$.

3 The World Model as an Optimal Estimation Problem

Definition 3.1 (Market world model). A *market world model* with parameters θ is a measurable map

$$\mathcal{M}_\theta : \mathcal{F}_t \times (0, \infty) \rightarrow \mathcal{P}(\mathcal{H})$$

such that $\mathcal{M}_\theta(\mathcal{F}_t, h) \approx \mathbb{P}(\mathbf{S}_{t+h} \in \cdot \mid \mathcal{F}_t)$.

Theorem 3.2 (Prediction error decomposition). For any $\mathcal{F}_t$ -measurable estimator $\hat{\mathbf{S}}_{t+h}$ of $\mathbf{S}_{t+h}$,

$$\mathbb{E} \left[\left\| \hat{\mathbf{S}}_{t+h} - \mathbf{S}_{t+h} \right\|_{\mathcal{H}}^2 \right] = \|\text{Bias}_\theta\|^2 + \underbrace{\text{Var}_\tau(h)}_{\text{physical}} + \underbrace{\text{Var}_\eta(h)}_{\text{behavioral}}, \quad (2)$$

where the last two terms are strictly positive and model-independent (see Theorem 4.4).

Proof. The bias-variance decomposition is standard. For the physical variance, write $\mathbf{S}_{t+h} = \hat{\mathbf{S}}_{t+h} + (\mathbf{S}_{t+h} - \mathbb{E}[\mathbf{S}_{t+h} \mid \mathcal{F}_t]) + (\mathbb{E}[\mathbf{S}_{t+h} \mid \mathcal{F}_t] - \hat{\mathbf{S}}_{t+h})$ and expand using $\mathbb{E}[M_t(A_t - B)] = 0$ for a martingale M and $\mathcal{F}_t$ -measurable B . The decomposition into physical and behavioral components follows from the independence established in Theorem 4.2. $\square$

4 Dual Stochastic Decomposition

4.1 Two Sources of Market Uncertainty

Standard models (e.g., Black–Scholes) conflate two qualitatively different sources of market randomness into a single Brownian motion. We argue this conflation is structurally incorrect:

- **Physical noise** τ_t : Arises from fundamental information asymmetry, parameter estimation error, and market microstructure frictions. It accumulates *continuously* along trading time—analogueous to thermal noise in physical systems.
- **Behavioral noise** η_t : Arises from investor sentiment, herding, panic selling, and euphoric buying. It manifests as *discontinuous jumps*—sudden regime shifts that cannot be generated by any continuous Brownian path.

Assumption 4.1 (Dual noise structure). The noise in equation (1) decomposes as:

$$\text{Noise}_t = \underbrace{\sigma_\tau(\mathbf{S}_t) dW_t^{(\tau)}}_{\text{physical}} + \underbrace{dJ_t^{(\eta)}}_{\text{behavioral}}, \quad (3)$$

where $W^{(\tau)}$ is a standard Brownian motion and $J_t^{(\eta)} = \int_0^t \int_{\mathbb{R}^k} z \tilde{N}^\eta(ds, dz)$ is an independent pure-jump Lévy process with Lévy measure ν^η satisfying $\int_{|z|>1} |z|^2 \nu^\eta(dz) < \infty$ and $[W^{(\tau)}, J^{(\eta)}]_t = 0$ a.s.

4.2 Canonical Decomposition Theorem

Theorem 4.2 (Dual Lévy–Itô decomposition). *Under Assumption 4.1, the total noise process $L_t := \sigma_\tau W_t^{(\tau)} + J_t^{(\eta)}$ has a unique Lévy characteristic (b, C, ν) that decomposes as:*

$$b = b_\tau + b_\eta, \quad C = \Sigma_\tau := \sigma_\tau \sigma_\tau^\top, \quad \nu = \nu^\eta, \quad (4)$$

with Σ_τ (the diffusion matrix) and ν^η (the jump intensity measure) uniquely identified by the characteristic function:

$$\mathbb{E}\left[e^{i\xi^\top L_t}\right] = \exp\left(-t\left[\frac{1}{2}\xi^\top \Sigma_\tau \xi + \int_{\mathbb{R}^k} \left(1 - e^{i\xi^\top z} + i\xi^\top z \mathbb{1}_{|z|\leq 1}\right) \nu^\eta(dz)\right]\right). \quad (5)$$

In particular, $\Sigma_\tau \neq 0$ and $\nu^\eta \neq 0$ are orthogonal in the sense that no jump measure can generate a continuous Gaussian component.

Proof. By independence of $W^{(\tau)}$ and $J^{(\eta)}$, the characteristic function of L_t factors as

$$\mathbb{E}[e^{i\xi^\top L_t}] = \mathbb{E}[e^{i\xi^\top \sigma_\tau W_t}] \cdot \mathbb{E}[e^{i\xi^\top J_t^\eta}] = e^{-t\Psi_\tau(\xi)} \cdot e^{-t\Psi_\eta(\xi)},$$

where $\Psi_\tau(\xi) = \frac{1}{2}\xi^\top \Sigma_\tau \xi$ (Gaussian) and $\Psi_\eta(\xi) = \int (1 - e^{i\xi^\top z} + i\xi^\top z \mathbb{1}_{|z| \leq 1}) \nu^\eta(dz)$ (pure-jump Lévy). Adding the two Lévy exponents gives (5). Uniqueness of the Lévy characteristic (Theorem 2.5) and the canonical decomposition of Lévy processes (Sato, 1999) establish that Σ_τ and ν^η are uniquely determined. The orthogonality statement reflects the Lévy–Khintchine classification: the Gaussian part is determined by $C = \Sigma_\tau$ alone, the jump part by ν^η alone; no assignment of mass to ν^η at any point in $\mathbb{R}^k$ can contribute to C . $\square$

Remark 4.3 (Heavy tails of behavioral noise). Empirical evidence (Cont and Bouchaud, 2000; Lux and Marchesi, 1999) supports $\nu^\eta(|z| > x) \sim x^{-\alpha}$ for $\alpha \in (1, 2)$ (heavy-tailed but with finite variance), consistent with the power-law tails observed in financial return distributions and inconsistent with any Gaussian model.

4.3 Prediction Lower Bound

Theorem 4.4 (Cramér–Rao-type prediction lower bound). *Let $\mathbf{S}$ satisfy (1) under Assumption 4.1. For any $\mathcal{F}_t$ -measurable unbiased estimator $\hat{\mathbf{S}}_{t+h}$ of $\mathbf{S}_{t+h}$ and any horizon $h > 0$:*

$$\text{Var}\left(\hat{\mathbf{S}}_{t+h} - \mathbf{S}_{t+h}\right) \geq \underbrace{\sigma_\tau^2 h}_{\text{physical lower bound}} + \underbrace{\lambda_\eta m_2^\eta h}_{\text{behavioral lower bound}} =: \text{CR}(h), \quad (6)$$

where $\sigma_\tau^2 = \text{tr}(\Sigma_\tau)$, $\lambda_\eta > 0$ is the jump intensity, and $m_2^\eta = \int_{\mathbb{R}^k} |z|^2 \nu^\eta(dz)$.

Proof. **Step 1 (Physical component).** Conditional on $\mathcal{F}_t$, the increment $\mathbf{S}_{t+h}^{(\tau)} - \mathbf{S}_t^{(\tau)} := \int_t^{t+h} \sigma_\tau dW_u$ is Gaussian with mean zero and covariance $\Sigma_\tau h$. For any unbiased estimator $\hat{A}$ of a Gaussian parameter with Fisher information I , the standard Cramér–Rao inequality gives $\text{Var}(\hat{A}) \geq I^{-1}$. Here $I^{-1} = \Sigma_\tau h$, so $\text{Var}(\hat{\mathbf{S}}_{t+h}^{(\tau)}) \geq \text{tr}(\Sigma_\tau)h = \sigma_\tau^2 h$.

Step 2 (Behavioral component). Let $K_{t,t+h} = N^\eta([t, t+h] \times \mathbb{R}^k)$ be the number of jumps in $[t, t+h]$. Given $K_{t,t+h} = n$, the jump sizes are i.i.d. with common distribution $\nu^\eta(dz)/\lambda_\eta$, so the total jump contribution has variance $n \cdot m_2^\eta / \lambda_\eta$. Integrating over $K_{t,t+h} \sim \text{Poisson}(\lambda_\eta h)$:

$$\text{Var}\left(\int_t^{t+h} \int z \tilde{N}^\eta(du, dz)\right) = \lambda_\eta h \cdot m_2^\eta.$$

Since $\hat{\mathbf{S}}_{t+h}$ cannot predict the future jump realisation without observing N^η beyond time t , this variance is irreducible.

Step 3 (Independence). By Assumption 4.1, physical and behavioral increments are independent, so their variances add: $\text{Var}(\mathbf{S}_{t+h}^{(\tau)} + J_{t+h}^{(\eta)} - J_t^{(\eta)}) = \sigma_\tau^2 h + \lambda_\eta m_2^\eta h = \text{CR}(h)$. $\square$

Remark 4.5 (Analogy with quantum mechanics). Theorem 4.4 is the financial analogue of the Heisenberg uncertainty principle: even with perfect knowledge of $\mathcal{F}_t$ and an optimal model, the sum of prediction variances across physical and behavioral channels cannot

fall below $\text{CR}(h)$. This bound is a structural feature of the market, not a limitation of methodology.

Corollary 4.6 (Time-varying temperature parameter). *The scalar temperature parameter τ_t of the world model (Ha and Schmidhuber, 2018) has the closed-form expression:*

$$\tau_t = \sqrt{\sigma_\tau^2(\mathbf{S}_t) + \lambda_\eta(t) m_2^\eta(t)},$$

and is a $\mathbb{F}$ -adapted, time-varying quantity determined by local noise intensities. Training the world model to match the observed local variance provides a principled, data-driven estimate of τ_t .

5 Financial Event Operator Algebra

5.1 Affine Event Operators

Definition 5.1 (Affine event operator). For each financial event type $w \in \mathcal{W}$, the associated *event operator* is the affine map:

$$T_w(s) = A_w s + b_w + \Sigma_w \varepsilon_w, \quad \varepsilon_w \sim \mathcal{N}(0, I), \quad (7)$$

where $A_w \in \mathbb{R}^{d' \times d}$ (with $d' = d$ or $d' \neq d$ depending on the event type), $b_w \in \mathbb{R}^{d'}$, and $\Sigma_w \in \mathbb{R}^{d' \times d'}$ encodes event-specific announcement uncertainty.

5.2 Classification of Financial Events

Financial events fall into three algebraic modes based on how they act on the state space:

Definition 5.2 (Three modes of action). (I) **Local endomorphism** ($d' = d$, acts on one asset): stock split, share repurchase, earnings announcement, dividend payment. $A_w \in \mathbb{R}^{d \times d}$; the state space dimension is preserved.

(II) **Global tensor action** (acts on all n assets via Kronecker product): central bank rate change, systemic shock, sector-wide regulation. $T_w^{\text{global}} = \Lambda_w \otimes I_d$, with $\Lambda_w \in \mathbb{R}^{n \times n}$ encoding heterogeneous sensitivities.

(III) **Pairwise morphism** ($d' \neq d$, changes entity count): merger & acquisition ($n \rightarrow n - 1$), spin-off / IPO ($n \rightarrow n + 1$). A_w is *non-square*; the state space dimension changes.

Proposition 5.3 (Information irreversibility). *For any Type-I event w with natural inverse w^{-1} (e.g., stock split followed by reverse split),*

$$T_{w^{-1}} \circ T_w = I + \mathcal{E}_w^{\text{info}},$$

where $\mathcal{E}_w^{\text{info}} \neq 0$ is an operator acting non-trivially on the information subspace $\{\iota\text{-coordinate}\}$ of the state vector. Even when A_w is invertible, the event releases information that cannot be unannounced: matrix invertibility does not imply physical reversibility.

Proof. Let $s = (p, v, \ell, \kappa, \iota)^\top$. For a stock split with ratio $r > 1$: $A_w = \text{diag}(1, 1, 1, r, 1)$, $b_w = 0$. The split also increases the disclosure level: $\iota \mapsto \min(\iota + \delta_w, 1)$ for some $\delta_w > 0$. Hence b_w has a positive entry in the ι -coordinate. The inverse operation $A_{w^{-1}} = \text{diag}(1, 1, 1, r^{-1}, 1)$ restores κ but leaves ι elevated, so $(T_{w^{-1}} \circ T_w)(s)_\iota = \iota + \delta_w \neq \iota$. $\square$

5.3 Groupoid Structure of the Event Algebra

Type-III events (mergers and spin-offs) change the state space dimension n , making the operator A_w non-square. This prevents the collection of all event operators from forming a semigroup (which requires a single carrier set). The correct algebraic structure is a *groupoid*.

Definition 5.4 (Financial event groupoid). Define the *financial event groupoid* $\mathcal{G}_{\text{fin}} = (\mathcal{W}, \circ, \text{src}, \text{tgt}, \text{id})$ by:

- **Objects:** $\text{Ob}(\mathcal{G}) = \{\mathcal{H}_S : S \text{ is a valid } n\text{-asset market configuration, } n \in \mathbb{N}\}$.
- **Morphisms:** For $T_w \in \text{Hom}(\mathcal{H}_S, \mathcal{H}_{S'})$, $\text{src}(T_w) = \mathcal{H}_S$ and $\text{tgt}(T_w) = \mathcal{H}_{S'}$.
- **Composition:** $T_{w_2} \circ T_{w_1}$ is defined if and only if $\text{src}(T_{w_2}) = \text{tgt}(T_{w_1})$.
- **Identities:** $\text{id}_{\mathcal{H}_S} = I_{d \cdot |S|}$.
- **Topology:** Endow each $\text{Hom}(\mathcal{H}_S, \mathcal{H}_{S'})$ with the operator norm topology; the whole space with the disjoint-union topology.

Theorem 5.5 (Groupoid structure). $\mathcal{G}_{\text{fin}}$ is a topological groupoid.

Proof. We verify the four groupoid axioms:

(i) *Well-defined composition.* Given $T_{w_1} : \mathcal{H}_S \rightarrow \mathcal{H}_{S'}$ and $T_{w_2} : \mathcal{H}_{S'} \rightarrow \mathcal{H}_{S''}$, their composition $T_{w_2} \circ T_{w_1} : \mathcal{H}_S \rightarrow \mathcal{H}_{S''}$ is the standard operator composition, which is well-defined in Banach spaces whenever the intermediate spaces match. The dimensionality compatibility condition $\text{src}(T_{w_2}) = \text{tgt}(T_{w_1})$ is exactly this matching requirement.

(ii) *Associativity.* Operator composition in Banach spaces is associative: $(T_{w_3} \circ T_{w_2}) \circ T_{w_1} = T_{w_3} \circ (T_{w_2} \circ T_{w_1})$ whenever both sides are defined.

(iii) *Identity morphisms.* For each $\mathcal{H}_S$, the identity operator $\text{id}_{\mathcal{H}_S} = I_{d|S|}$ satisfies $T_w \circ \text{id}_{\mathcal{H}_S} = T_w$ and $\text{id}_{\mathcal{H}_{S'}} \circ T_w = T_w$ for any $T_w \in \text{Hom}(\mathcal{H}_S, \mathcal{H}_{S'})$.

(iv) *Inverses (partial).* For Type-I and Type-II events with invertible A_w (i.e., $\det(A_w) \neq 0$), the inverse T_w^{-1} exists as an operator and $T_w \circ T_w^{-1} = \text{id}_{\mathcal{H}_S}$ (up to the information residual in Proposition 5.3). For Type-III events (non-square A_w), the Moore–Penrose

pseudoinverse T_w^+ satisfies $T_w T_w^+ T_w = T_w$ but is not a full inverse—this is the natural partial-inverse structure of a groupoid.

(v) *Continuity.* Under the operator norm topology, $(T_{w_1}, T_{w_2}) \mapsto T_{w_2} \circ T_{w_1}$ is continuous (composition of bounded operators is jointly continuous in the norm topology). Similarly, $T_w \mapsto T_w^+$ is continuous on the set of operators with fixed rank. Hence $\mathcal{G}_{\text{fin}}$ is a topological groupoid. $\square$

Corollary 5.6 (Endomorphism monoid as a sub-structure). *The collection of Type-I events $\mathcal{W}_I \subset \mathcal{W}$ acting on a fixed n -asset space forms an endomorphism monoid with identity I_{5n} . For Type-I events with $\det(A_w) \neq 0$ this monoid is a group.*

6 The E-Game-C Architecture

6.1 Limitations of the Original VMC World Model

The V–M–C architecture of Ha and Schmidhuber (2018) consists of: a Variational AutoEncoder (**V**) that compresses high-dimensional observations to a latent code z_t ; a recurrent world model (**M**) that predicts z_{t+1} from (z_t, a_t) ; and a linear controller (**C**) trained by evolution strategies.

Direct application to finance faces three structural obstacles: (a) the input is heterogeneous (prices, news, event sequences), not pixels; (b) the “environment” is not fixed—it is the aggregate outcome of all agents’ decisions, creating a reflexive feedback loop; (c) the world model module M implicitly assumes the environment is a fixed dynamical system, ignoring the multi-agent strategic structure that is fundamental to financial markets.

6.2 Encoder Module E

Definition 6.1 (Information encoder). The encoder $\mathcal{E}_\phi : \mathcal{F}_t \rightarrow \mathcal{Z}$ maps the information set at time t to a latent code $z_t \in \mathcal{Z} \subset \mathbb{R}^{d_z}$ via a variational autoencoder (Kingma and Welling, 2014):

$$z_t = \mathcal{E}_\phi(\mathbf{x}_t), \quad \mathbf{x}_t = (p_t, v_t, \text{news}_t, \text{events}_t, \dots) \in \mathcal{F}_t,$$

trained by maximising the ELBO:

$$\mathcal{L}_{\text{ELBO}}(\phi, \psi) = \mathbb{E}_{z \sim q_\phi} [\log p_\psi(\mathbf{x}|z)] - D_{\text{KL}}(q_\phi(z|\mathbf{x}) \| p(z)).$$

6.3 Game Module: Mean-Field Equilibrium Operator

The core innovation of our architecture is replacing the recurrent world model M with a mean-field equilibrium operator.

Definition 6.2 (Market mean-field game). The market consists of a continuum of agents $u \in [0, 1]$. Agent u 's latent state evolves as:

$$dz_t^u = f(z_t^u, a_t^u, \mu_t, u) dt + \sigma dW_t^u, \quad z_0^u \sim \rho_0, \quad (8)$$

where $a_t^u \in \mathcal{A}$ is the action, $\mu_t = \text{Law}(z_t^u)$ is the *mean field* (distribution of all agents), and $(W^u)_{u \in [0, 1]}$ are independent Brownian motions. Agent u minimises the cost:

$$J^u(a) = \mathbb{E} \left[\int_0^T \ell(z_t^u, a_t^u, \mu_t, u) dt + g(z_T^u, \mu_T, u) \right]. \quad (9)$$

Theorem 6.3 (MFG equilibrium existence). *Under the following conditions:*

(A1) *f and ℓ are jointly continuous in all arguments and Lipschitz in (z, a, μ) with constant L_f ;*

(A2) *The action space $\mathcal{A}$ is compact and convex;*

(A3) *There exists a unique minimiser $\hat{a}(z, \mu, u) = \arg \min_{a \in \mathcal{A}} \mathcal{H}(z, a, \mu, u, p)$ for the Hamiltonian $\mathcal{H}$ with adjoint p ;*

(A4) (Lasry–Lions monotonicity) *$\int [\ell(z, \mu, a) - \ell(z, \mu', a)] d(\mu - \mu')(z) \geq 0$ for all μ, μ', a ,*

there exists a unique mean-field game equilibrium $(\hat{a}^, \hat{\mu})$ satisfying: (i) $\hat{a}_t^* = \arg \min_a J^u(a | \hat{\mu})$ for μ -a.e. u ; (ii) $\hat{\mu}_t = \text{Law}(z_t^u | \hat{a}^*)$ for all t .*

Proof. Define the fixed-point map $\Phi : C([0, T]; \mathcal{P}_2(\mathcal{Z})) \rightarrow C([0, T]; \mathcal{P}_2(\mathcal{Z}))$ by: given a flow $\mathbf{m} = (\mu_t)$, solve the forward SDE (8) with optimal feedback $\hat{a}(z, \mu_t, u)$ and output $\Phi(\mathbf{m})_t = \text{Law}(z_t^u)$.

Self-mapping. By (A1)–(A2) and Gronwall, solutions of (8) satisfy a uniform L^2 bound, so $\Phi(\mathbf{m}) \in C([0, T]; \mathcal{P}_2(\mathcal{Z}))$.

Compactness. Tightness of $\{\Phi(\mathbf{m}_n)\}$ for bounded sequences $\{\mathbf{m}_n\}$ follows from Arzela–Ascoli applied to the SDE solutions.

Continuity. If $\mathbf{m}_n \rightarrow \mathbf{m}$ in $C([0, T]; \mathcal{P}_2(\mathcal{Z}))$ (Wasserstein-2 topology), then by (A1) and stability of SDEs, $\Phi(\mathbf{m}_n) \rightarrow \Phi(\mathbf{m})$.

Schauder's theorem. Φ maps the closed convex set $\mathcal{K} = \{\mathbf{m} : \sup_t W_2(\mu_t, \rho_0)^2 \leq R\}$ (for R large enough) into itself, continuously, and $\mathcal{K}$ is compact. Schauder's fixed-point theorem (Evans, 2010) yields a fixed point $\hat{\mathbf{m}}$ with $\Phi(\hat{\mathbf{m}}) = \hat{\mathbf{m}}$, i.e., a MFG equilibrium.

Uniqueness. The Lasry–Lions condition (A4) ensures that any two equilibria $(\hat{a}^*, \hat{\mu})$ and $(\hat{a}^{**}, \hat{\mu}^{**})$ satisfy $W_2(\hat{\mu}_t, \hat{\mu}_t^{**}) = 0$ for all t (by a standard energy estimate using the monotonicity of ℓ), hence uniqueness. $\square$

6.4 Controller Module C

Definition 6.4 (Optimal controller). The controller $\pi^* : \mathcal{Z} \times \mathcal{H}_{\text{RNN}} \rightarrow \mathcal{A}$ maps the current latent state z_t and world-model hidden state h_t to an optimal action:

$$\pi^*(z_t, h_t) = \arg \max_{a \in \mathcal{A}} Q^*(z_t, h_t, a),$$

where Q^* is the optimal action-value function satisfying the HJB equation:

$$0 = \partial_t Q + \sup_{a \in \mathcal{A}} [f(z, a, \hat{\mu}_t)^\top \nabla_z Q + \ell(z, a, \hat{\mu}_t)] + \frac{1}{2} \sigma^2 \Delta_z Q.$$

7 Hierarchical Mean-Field Theory

7.1 Motivation: Multi-Scale Structure of Financial Markets

A key structural observation motivates the hierarchical extension:

1. At the **national/regional scale**, sovereign entities (countries, central banks, regulatory bodies) interact without a supranational central planner—a MFG structure.
2. Within each nation, individual firms are subject to regulation and monetary policy set by the central bank—a partial MFC structure.
3. At the **firm level**, individual investors within a firm’s investor base again interact without coordination—a MFG structure.

This hierarchical interplay—MFG at the macro level, mixed MFG/MFC at the meso level—cannot be captured by a single-level mean-field model.

7.2 Two-Level McKean–Vlasov System

Let $N \in \mathbb{N}$ denote the number of groups (nations) and M the size of the continuum of agents within each group.

Level 1 (macro-group dynamics). Each group $j \in \{1, \dots, N\}$ has a representative state $\xi_t^j \in \mathbb{R}^{d_1}$ evolving as:

$$d\xi_t^j = b_1(\xi_t^j, \nu_t^{(1)}, \alpha_t^j) dt + \sigma_1 dB_t^j, \quad \xi_0^j = \xi_0^j \in \mathbb{R}^{d_1}, \quad (10)$$

where $\nu_t^{(1)} = \frac{1}{N} \sum_{k=1}^N \delta_{\xi_t^k}$ is the empirical measure of macro-group states, α_t^j is the policy (control) of group j , and $(B^j)_{j=1}^N$ are independent Brownian motions. Group j minimises the macro-cost:

$$J_1^j(\alpha^j) = \mathbb{E} \left[\int_0^T f_1(\xi_t^j, \nu_t^{(1)}, \alpha_t^j) dt + g_1(\xi_T^j, \nu_T^{(1)}) \right]. \quad (11)$$

Level 2 (micro-agent dynamics). Within group j , a continuum of agents $u \in [0, 1]$ has states $x_t^{j,u} \in \mathbb{R}^{d_2}$ evolving as:

$$dx_t^{j,u} = b_2(x_t^{j,u}, \mu_{j,t}^{(2)}, \xi_t^j, a_t^{j,u}) dt + \sigma_2 dW_t^{j,u}, \quad (12)$$

where $\mu_{j,t}^{(2)} = \int_0^1 \delta_{x_t^{j,u}} du$ is the intra-group mean field, $a_t^{j,u}$ is agent u 's control, and $(W^{j,u})$ are i.i.d. Each agent minimises:

$$J_2^{j,u}(a) = \mathbb{E} \left[\int_0^T f_2(x_t^{j,u}, \mu_{j,t}^{(2)}, \xi_t^j, a_t^{j,u}) dt + g_2(x_T^{j,u}, \mu_{j,T}^{(2)}, \xi_T^j) \right]. \quad (13)$$

Coupling condition. The Level-1 and Level-2 dynamics are coupled through the *aggregation functional*

$$\Psi_j(\mu_{j,\cdot}^{(2)}) := \int_{\mathbb{R}^{d_2}} \phi(x) \mu_{j,t}^{(2)}(dx), \quad t \in [0, T], \quad (14)$$

for a bounded Lipschitz test function ϕ ; the macro-group cost J_1^j depends on $\Psi_j(\mu_{j,\cdot}^{(2)})$ through the drift b_1 .

7.3 Hierarchical Nash Equilibrium

Definition 7.1 (Hierarchical Nash equilibrium). A pair $((\alpha^{j,*})_{j=1}^N, (a^{j,u,*})_{j,u})$ is a *hierarchical Nash equilibrium* if:

- (i) For each group j and the resulting Level-2 equilibrium $\hat{\mu}_j^{(2)}[\xi^j]$ (defined in Definition 7.5 below), the macro policy satisfies:

$$J_1^j(\alpha^{j,*}) \leq J_1^j(\alpha^j) \quad \forall \alpha^j, \text{ given } (\alpha^{k,*})_{k \neq j};$$

- (ii) For each group j and each agent $u \in [0, 1]$, given ξ^j and $\mu_{j,t}^{(2)} = \hat{\mu}_{j,t}^{(2)}[\xi^j]$:

$$J_2^{j,u}(a^{j,u,*}) \leq J_2^{j,u}(a^{j,u}) \quad \forall a^{j,u}.$$

7.4 Assumptions

Assumption 7.2 (Regularity at Level 2). (L1) b_2, f_2, g_2 are jointly continuous and Lipschitz in (x, μ, ξ, a) with constant $L_2 < \infty$, uniformly in j .

(L2) b_2 satisfies linear growth: $|b_2(x, \mu, \xi, a)| \leq C_2(1 + |x|)$.

(L3) The action space $\mathcal{A}_2$ is compact and convex.

(L4) Lasry–Lions monotonicity at Level 2: $\int [f_2(x, \mu, \xi, a) - f_2(x, \mu', \xi, a)] d(\mu - \mu')(x) \geq 0$.

Assumption 7.3 (Regularity at Level 1). (L5) b_1, f_1, g_1 are jointly continuous and Lipschitz in (ξ, ν, α) with constant $L_1 < \infty$.

(L6) The action space $\mathcal{A}_1$ is compact and convex.

(L7) Lasry–Lions monotonicity at Level 1: $\int [f_1(\xi, \nu, \alpha) - f_1(\xi, \nu', \alpha)] d(\nu - \nu')(\xi) \geq 0$.

(L8) The aggregation map $\Psi_j : C([0, T]; \mathcal{P}_2(\mathbb{R}^{d_2})) \rightarrow C([0, T]; \mathbb{R})$ is Lipschitz with constant $L_\Psi < \infty$.

7.5 Main Theorem

Theorem 7.4 (Hierarchical Mean-Field Nash Equilibrium). *Under Assumptions 7.2 and 7.3, there exists a unique hierarchical Nash equilibrium in the sense of Definition 7.1.*

The proof proceeds through a sequence of four lemmas.

Definition 7.5 (Level-2 equilibrium operator). For each fixed continuous path $\xi = (\xi_t)_{t \in [0, T]} \in C([0, T]; \mathbb{R}^{d_1})$, let $\text{MFG}_2[\xi]$ denote the mean-field game with dynamics (12)–(13) with ξ_t^j replaced by ξ_t . By Theorem 6.3 (applied to the Level-2 system under (L1)–(L4)), $\text{MFG}_2[\xi]$ has a unique equilibrium measure flow $\hat{\mu}^{(2)}[\xi] \in C([0, T]; \mathcal{P}_2(\mathbb{R}^{d_2}))$.

Lemma 7.6 (Level-2 stability estimate). *Under Assumption 7.2, for any two paths $\xi, \xi' \in C([0, T]; \mathbb{R}^{d_1})$:*

$$\sup_{t \in [0, T]} W_2(\hat{\mu}_t^{(2)}[\xi], \hat{\mu}_t^{(2)}[\xi']) \leq C_{2, \text{stab}} \|\xi - \xi'\|_{C([0, T])},$$

where $C_{2, \text{stab}} = C_{2, \text{stab}}(L_2, T, \sigma_2) < \infty$ depends only on the problem parameters.

Proof. Let x_t and x'_t be the optimal state processes for $\text{MFG}_2[\xi]$ and $\text{MFG}_2[\xi']$ respectively. Define $\Delta_t = x_t - x'_t$. By the Lipschitz condition (L1), the dynamics of Δ_t satisfy:

$$d\Delta_t = [b_2(x_t, \hat{\mu}_t, \xi_t, \hat{a}_t) - b_2(x'_t, \hat{\mu}'_t, \xi'_t, \hat{a}'_t)] dt.$$

Using the triangle inequality for the Lipschitz bound:

$$|d\Delta_t| \leq L_2(|\Delta_t| + W_2(\hat{\mu}_t, \hat{\mu}'_t) + |\xi_t - \xi'_t| + |\hat{a}_t - \hat{a}'_t|) dt.$$

Since the optimal control satisfies the same Lipschitz bound (by the smoothness of the Hamiltonian), and $W_2(\hat{\mu}_t, \hat{\mu}'_t) \leq (\mathbb{E}[|\Delta_t|^2])^{1/2}$, Gronwall's inequality gives:

$$\mathbb{E} \left[\sup_{t \leq T} |\Delta_t|^2 \right] \leq C e^{C L_2 T} \|\xi - \xi'\|_{C([0, T])}^2.$$

Since $\hat{\mu}_t^{(2)}[\xi] = \text{Law}(x_t)$ and $\hat{\mu}_t^{(2)}[\xi'] = \text{Law}(x'_t)$, the W_2 distance is bounded by $(\mathbb{E}[|\Delta_t|^2])^{1/2}$, yielding the claim. $\square$

Lemma 7.7 (Continuity of the aggregation functional). *The composed map $\xi \mapsto \Psi_j(\hat{\mu}^{(2)}[\xi])$ from $C([0, T]; \mathbb{R}^{d_1})$ to $C([0, T]; \mathbb{R})$ is Lipschitz with constant $L_\Psi \cdot C_{2, \text{stab}}$.*

Proof. Direct application of (L8) and Lemma 7.6:

$$\begin{aligned} \|\Psi_j(\hat{\mu}^{(2)}[\xi]) - \Psi_j(\hat{\mu}^{(2)}[\xi'])\|_{C([0, T])} &\leq L_\Psi \sup_t W_2(\hat{\mu}_t^{(2)}[\xi], \hat{\mu}_t^{(2)}[\xi']) \\ &\leq L_\Psi C_{2, \text{stab}} \|\xi - \xi'\|_{C([0, T])}. \end{aligned}$$

$\square$

Lemma 7.8 (Existence of Level-1 MFG equilibrium). *Fix the Level-2 equilibrium measure flows $\hat{\mu}^{(2)}[\xi^j]$ as a function of the Level-1 paths ξ^j . Then the Level-1 system (10)–(11) (with the aggregation term incorporated into b_1) admits a unique MFG equilibrium $(\hat{\alpha}^{j,*}, \hat{\nu}^{(1)})$.*

Proof. The Level-1 system is a standard mean-field game with state space $\mathbb{R}^{d_1}$ and action space $\mathcal{A}_1$. By Lemma 7.7, incorporating $\Psi_j(\hat{\mu}^{(2)}[\xi^j])$ into the Level-1 drift b_1 produces a new drift $\tilde{b}_1(\xi^j, \nu^{(1)}, \alpha^j) := b_1(\xi^j, \nu^{(1)}, \alpha^j, \Psi_j(\hat{\mu}^{(2)}[\xi^j]))$ that is Lipschitz in $(\xi^j, \nu^{(1)}, \alpha^j)$ with constant $\max(L_1, L_\Psi C_{2, \text{stab}})$. Assumptions (L5)–(L7) ensure that Theorem 6.3 applies to the Level-1 system with this modified drift, giving the existence and uniqueness of $(\hat{\alpha}^{j,*}, \hat{\nu}^{(1)})$. $\square$

Lemma 7.9 (Consistency: Level-2 under optimal Level-1 policies). *Given the Level-1 equilibrium trajectories $(\hat{\xi}_t^j)$ generated by $\hat{\alpha}^{j,*}$, the corresponding Level-2 equilibria $\hat{a}^{j,u,*} = \hat{a}^{j,u,*}[\hat{\xi}^j]$ satisfy the Level-2 optimality conditions in Definition 7.1(ii).*

Proof. By Definition 7.5, $\hat{a}^{j,u,*}[\hat{\xi}^j]$ is the equilibrium of $\text{MFG}_2[\hat{\xi}^j]$, which directly gives optimality. No further argument is needed beyond the definition. $\square$

Proof of Theorem 7.4. Existence.

1. By Theorem 6.3 applied at Level 2 (Lemmas exist for any fixed ξ^j), the Level-2 equilibrium operator $\xi^j \mapsto (\hat{a}^{j,u,*}[\xi^j], \hat{\mu}_j^{(2)}[\xi^j])$ is well-defined.
2. By Lemma 7.6 and 7.7, the aggregation functional $\xi^j \mapsto \Psi_j(\hat{\mu}_j^{(2)}[\xi^j])$ is Lipschitz.
3. By Lemma 7.8, the Level-1 MFG with the aggregated feedback has a unique equilibrium $(\hat{\alpha}^{j,*}, \hat{\nu}^{(1)})$, generating equilibrium trajectories $\hat{\xi}_t^j$.
4. By Lemma 7.9, the Level-2 policies $\hat{a}^{j,u,*}[\hat{\xi}^j]$ are optimal given $\hat{\xi}^j$. Together with the Level-1 optimality from step 3, Definition 7.1 is satisfied.

Uniqueness. Suppose $((\alpha^{j,*}), (a^{j,u,*}))$ and $((\alpha^{j,**}), (a^{j,u,**}))$ are two hierarchical Nash equilibria generating trajectory distributions $\nu^{(1)}$ and $\nu^{(1),**}$ respectively.

At Level 2: by the uniqueness part of Theorem 6.3 (uniqueness follows from the Lasry–Lions condition (L4)), for each fixed Level-1 trajectory path ξ^j : $\hat{\mu}_j^{(2)}[\xi^j]$ is unique, so the Level-2 equilibria coincide whenever the Level-1 paths coincide.

At Level 1: the Level-1 system with the (unique) aggregation $\Psi_j(\hat{\mu}_j^{(2)}[\xi^j])$ is a standard MFG satisfying (L5)–(L7), whose unique equilibrium measure flow $\hat{\nu}^{(1)}$ is guaranteed by the Lasry–Lions condition (L7). Hence $\nu^{(1)} = \nu^{(1),**}$, and the Level-1 equilibrium trajectories coincide. Feeding these back into the Level-2 uniqueness gives $a^{j,u,*} = a^{j,u,**}$ μ_j -a.e. $\square$

Corollary 7.10 (Multi-scale information decomposition). *Under the hierarchical equilibrium, the optimal micro-agent policy decomposes:*

$$a_t^{j,u,*} = \underbrace{\varphi_1(\hat{\xi}_t^j, \hat{\nu}_t^{(1)})}_{\text{macro contribution}} + \underbrace{\varphi_2(x_t^{j,u}, \hat{\mu}_{j,t}^{(2)}, \hat{\xi}_t^j)}_{\text{micro contribution}} + O(\epsilon),$$

where $\epsilon = L_\Psi C_{2,\text{stab}} \cdot \sup_t |\xi_t^j - \hat{\xi}_t^j|$ measures the coupling strength.

Proof. Expand the optimal value function $V^{j,u}(x, \mu, \xi)$ around the equilibrium $(\hat{\mu}_{j,t}^{(2)}, \hat{\xi}_t^j)$ using Taylor’s theorem. The first-order terms in ξ yield φ_1 via the envelope theorem, and the terms in μ yield φ_2 . The $O(\epsilon)$ remainder follows from Lemma 7.6. $\square$

8 Stochastic Lyapunov Stability

8.1 Lyapunov Stability under Dual Noise

The classical Lyapunov stability theory for ODEs must be substantially extended when the noise includes a jump component. The key tool is the Itô formula for Lévy-driven processes.

Definition 8.1 (Infinitesimal generator). For the process $\mathbf{S}_t$ satisfying (1) (ignoring event jumps for clarity), the *extended generator* acting on $C^2(\mathcal{H})$ is:

$$\begin{aligned} \mathcal{L}V(\mathbf{S}) &= \nabla V(\mathbf{S})^\top \mu(\mathbf{S}) + \frac{1}{2} \text{tr}(\sigma_\tau \sigma_\tau^\top \nabla^2 V(\mathbf{S})) \\ &\quad + \int_{\mathbb{R}^k} [V(\mathbf{S} + z) - V(\mathbf{S}) - z^\top \nabla V(\mathbf{S}) \mathbb{1}_{|z| \leq 1}] \nu^\eta(dz). \end{aligned} \quad (15)$$

The three terms correspond respectively to: the contribution of the deterministic drift, the diffusive physical noise, and the behavioral jump noise.

Theorem 8.2 (Stochastic Lyapunov stability and invariant measure). *Suppose there exists a C^2 function $V : \mathcal{H} \rightarrow \mathbb{R}_+$ and constants $c, c_0 > 0$ such that:*

$$\mathcal{L}V(\mathbf{S}) \leq -cV(\mathbf{S}) + c_0 \quad \text{for all } \mathbf{S} \in \mathcal{H}. \quad (16)$$

Then:

(i) **Exponential ergodicity:** *For any initial condition $\mathbf{S}_0$,*

$$\mathbb{E}[V(\mathbf{S}_t)] \leq V(\mathbf{S}_0) e^{-ct} + \frac{c_0}{c}.$$

(ii) **Existence of invariant measure:** *The process $\mathbf{S}_t$ admits at least one invariant (stationary) probability measure π^* .*

(iii) **Exponential convergence:** *If additionally $V \geq 1$ and the sublevel sets $\{V \leq r\}$ are compact (petite sets for the Markov chain skeleton), then π^* is unique and*

$$\|\mathbb{P}(\mathbf{S}_t \in \cdot | \mathbf{S}_0) - \pi^*\|_{\text{TV}} \leq C(\mathbf{S}_0) e^{-\rho t}$$

for some $C(\mathbf{S}_0) < \infty$ and $\rho > 0$ depending on c and c_0 .

Proof. (i) *Exponential bound.* By Itô's formula for Lévy processes (Sato, 1999; Protter, 2005):

$$\begin{aligned} dV(\mathbf{S}_t) &= \mathcal{L}V(\mathbf{S}_t) dt + \nabla V(\mathbf{S}_t)^\top \sigma_\tau dW_t \\ &\quad + \int_{\mathbb{R}^k} [V(\mathbf{S}_{t-} + z) - V(\mathbf{S}_{t-})] \tilde{N}^n(dt, dz). \end{aligned}$$

The stochastic integrals are local martingales. Taking expectations:

$$\frac{d}{dt} \mathbb{E}[V(\mathbf{S}_t)] = \mathbb{E}[\mathcal{L}V(\mathbf{S}_t)] \leq -c \mathbb{E}[V(\mathbf{S}_t)] + c_0.$$

Gronwall's inequality gives $\mathbb{E}[V(\mathbf{S}_t)] \leq V(\mathbf{S}_0) e^{-ct} + c_0/c$.

(ii) *Existence of invariant measure.* The bound in (i) shows that the family $\{(1/t) \int_0^t \mathbb{P}(\mathbf{S}_s \in \cdot) ds\}_{t>0}$ is tight in $\mathcal{P}(\mathcal{H})$ (by Markov's inequality and the upper bound c_0/c on $\mathbb{E}[V]$). Any weak limit point of this family is an invariant measure (Krylov–Bogoliubov theorem (Da Prato and Zabczyk, 1996)).

(iii) *Uniqueness and exponential rate.* Under the petite-set condition, the process is *V-geometrically ergodic* in the sense of Meyn and Tweedie (1993) (their Foster–Lyapunov Criterion V). The uniqueness of π^* and the TV convergence follow from their Theorem 15.0.1. $\square$

Remark 8.3 (Lyapunov condition in financial terms). The condition (16) with $V(\mathbf{S}) = \|\mathbf{S} - \mathbf{S}^*\|^2$ reduces to a mean-reversion condition: the drift and jump terms together must

push the market state back toward $\mathbf{S}^*$ at rate at least c , i.e.,

$$2(\mathbf{S} - \mathbf{S}^*)^\top \mu(\mathbf{S}) + \text{tr}(\sigma_\tau \sigma_\tau^\top) + \int_{\mathbb{R}^k} (2(\mathbf{S} - \mathbf{S}^*)^\top z + |z|^2) \nu^\eta(dz) \leq -c\|\mathbf{S} - \mathbf{S}^*\|^2 + c_0.$$

8.2 Bubbles and Crises as Lyapunov Violations

Definition 8.4 (Instability region). The *instability region* is $\mathcal{R}_{\text{unstable}} := \{\mathbf{S} \in \mathcal{H} : \mathcal{L}V(\mathbf{S}) > 0\}$.

Proposition 8.5 (Analytic signature of market bubbles). *If $\mathbf{S}_0 \in \mathcal{R}_{\text{unstable}}$ and the behavioral jump intensity $\lambda_\eta(t)$ is sufficiently large (high-sentiment regime), then:*

$$\mathbb{P}(\tau_{\text{escape}} < T) > 0, \quad \tau_{\text{escape}} := \inf\{t > 0 : V(\mathbf{S}_t) > R\}$$

for any $R > V(\mathbf{S}_0)$. That is, the market state escapes to a high- V region (bubble or crash trajectory) with positive probability.

Proof. In $\mathcal{R}_{\text{unstable}}$, $\mathcal{L}V > 0$, so $\mathbb{E}[V(\mathbf{S}_{t \wedge \tau})]$ is locally increasing. By the optional stopping theorem applied to the stopped submartingale $V(\mathbf{S}_{t \wedge \tau})$, the probability of exiting $\{V \leq R\}$ before time T is bounded below by $\mathbb{P}(\tau_{\text{escape}} < T) \geq \mathcal{L}V(\mathbf{S}_0)/(R - V(\mathbf{S}_0)) \cdot \delta + O(\delta^2)$ for small $\delta > 0$ (first-exit probability estimate). $\square$

Corollary 8.6 (Real-time market risk indicator). *Define the instantaneous risk indicator:*

$$\text{RI}(t) := \mathcal{L}V(\mathbf{S}_t).$$

$\text{RI}(t) > 0$ signals that the market is in a potential bubble or crisis regime; $\text{RI}(t) < -cV(\mathbf{S}_t)$ signals strong mean-reversion. Monitoring RI provides a theoretically grounded, model-consistent early-warning indicator.

9 Unified Evolution Equation

We now assemble all components into the complete evolution equation.

Theorem 9.1 (Unified market world model equation). *The market state process $\mathbf{S}_t$ under the E-Game-C architecture, with dual noise and financial event operators, satisfies:*

$$\mathbf{S}_t = \mathbf{S}_0 + \int_0^t \mu^*(\mathbf{S}_u, \hat{\mu}_u) du + \int_0^t \sigma_\tau(\mathbf{S}_u) dW_u^{(\tau)} + \int_0^t \int_{\mathbb{R}^k} z \tilde{N}^\eta(du, dz) + \sum_{\substack{w \in \mathcal{W}_I \cup \mathcal{W}_{II} \\ \tau_w \leq t}} (T_w - I) \mathbf{S}_{\tau_w^-} + \sum_{\substack{w \in \mathcal{W}_{III} \\ \tau_w \leq t}} \mathcal{R}_w(\mathbf{S}_{\tau_w^-}) \quad (17)$$

where $\mu^*(\mathbf{S}, \hat{\mu})$ is the equilibrium drift from the hierarchical MFG (Theorem 7.4), T_w are Type-I/II event operators, and $\mathcal{R}_w$ are Type-III (dimension-changing) groupoid morphisms.

Proof. The equation follows by assembling: (a) the Itô integral form (Definition 2.2); (b) the dual noise decomposition (Theorem 4.2); (c) the event operator jumps (Definition 5.1); (d) the equilibrium drift from the hierarchical MFG equilibrium (Theorem 7.4), which replaces the generic $\mu(\mathbf{S}, u)$. Well-posedness of (17) follows from the Lipschitz conditions in Assumptions 7.2–7.3 and the standard existence theory for McKean–Vlasov SDEs with jumps (Méléard, 1996). $\square$

10 Discussion

10.1 Relation to Prior Work

Factor models. Our framework subsumes factor models as a special case: setting $\mu^* = \sum_k \beta_k F_k$ (linear factor structure), $\sigma_\tau = \sigma_0 I$, $\nu^\eta = 0$, and $T_w = I$ (no events) recovers a multi-factor APT model. The advantage of our setting is that it does not assume stationarity of factor loadings.

MFG in finance. Existing MFG-based financial models (Carmona and Lacker, 2015; Carmona and Mouzouni, 2023) operate at a single scale. Our hierarchical extension (Theorem 7.4) is, to our knowledge, the first to prove existence of a hierarchical Nash equilibrium in a two-level McKean–Vlasov system with explicit Level-1/Level-2 coupling via an aggregation functional.

World models. The world-model paradigm has been applied to portfolio optimisation (AWMLG, 2026) and order-book simulation (JPMorgan, 2019) but without a theoretical foundation. Our E-Game-C architecture provides this foundation and subsumes prior architectures as special cases (e.g., Ha and Schmidhuber (2018) corresponds to a Level-1-only model with no behavioral noise).

10.2 Limitations and Future Work

Empirical validation. This paper establishes the theoretical framework. Empirical validation—calibrating (σ_τ, ν^η) from high-frequency data, estimating $\hat{\mu}^{(2)}$ from a public dataset, testing the Lyapunov risk indicator—is deferred to subsequent work.

Graphon extension. The agent type u is currently a scalar in $[0, 1]$. As noted in Section 7, the natural extension is $u \in [0, 1]^m$ (a vector of agent characteristics), leading to Graphon mean-field games (Carmona et al., 2022). This extension is theoretically straightforward given our framework but requires additional functional-analytic machinery.

Non-Markovian extensions. The Markovian assumption (McKean–Vlasov SDE) can be relaxed to allow path-dependent coefficients, at the cost of working in infinite-dimensional path spaces.

11 Conclusion

We have presented the first systematic mathematical theory for market world models. Starting from the observation that financial markets have no correct ODE description (Proposition 2.1), we developed:

- A dual noise decomposition into physical (Brownian) and behavioral (Lévy-jump) components, with a Cramér–Rao lower bound (Theorem 4.4) that quantifies the irreducible prediction uncertainty of any world model.
- A groupoid algebra of financial event operators (Theorem 5.5), providing the first algebraically rigorous treatment of corporate events as structured transformations of market state.
- A hierarchical mean-field theory with a complete existence and uniqueness proof of the hierarchical Nash equilibrium (Theorem 7.4), capturing the simultaneous presence of MFG and MFC structures at different market scales.
- A stochastic Lyapunov stability theorem under combined Brownian–Lévy noise (Theorem 8.2), yielding a theoretically grounded real-time market risk indicator.

The E-Game-C architecture ties these components into an executable framework whose equilibrium drift is provided by the hierarchical MFG and whose noise structure is governed by the dual decomposition.

Together, these results place the world-model paradigm for quantitative finance on a rigorous mathematical foundation comparable to the classical Itô-calculus foundations of derivative pricing.

A Technical Lemmas

Lemma A.1 (Gronwall’s inequality, integral form). *If $u(t) \leq \alpha(t) + \int_0^t \beta(s) u(s) ds$ with α non-decreasing and $\beta \geq 0$, then $u(t) \leq \alpha(t) \exp \int_0^t \beta(s) ds$.*

Lemma A.2 (W_2 distance and optimal coupling). *For probability measures μ, μ' on $\mathbb{R}^d$, $W_2(\mu, \mu')^2 = \inf_{\gamma \in \Gamma(\mu, \mu')} \int |x - y|^2 d\gamma(x, y)$, and for measures arising as laws of random variables X, X' : $W_2(\text{Law}(X), \text{Law}(X')) \leq (\mathbb{E}[|X - X'|^2])^{1/2}$.*

B Proof of Non-Commutativity of Event Sequences

Proposition B.1 (Event sequences are non-commutative). *In general, $T_{w_2} \circ T_{w_1} \neq T_{w_1} \circ T_{w_2}$.*

Proof. Take the single-asset state $s = (p, \kappa)^\top \in \mathbb{R}^2$ (log-price and log-shares). Let w_1 be a 2-for-1 stock split: $T_{w_1}(s) = (p, \kappa + \log 2)^\top$. Let w_2 be a cash dividend of d per share: $T_{w_2}(s) = (p - \log(1 + d/e^p), \kappa)^\top$ (ex-dividend price drop).

Then: $(T_{w_2} \circ T_{w_1})(s) = T_{w_2}(p, \kappa + \log 2) = (p - \log(1 + d/e^p), \kappa + \log 2)$, while $(T_{w_1} \circ T_{w_2})(s) = T_{w_1}(p - \log(1 + d/e^p), \kappa) = (p - \log(1 + d/e^p), \kappa + \log 2)$. In this case they coincide. However, if w_2 is instead a rights issue that changes κ : $T_{w_2}(s) = (p \cdot r^{-1}, \kappa + \log r)^\top$, then $T_{w_2} \circ T_{w_1}(s) = T_{w_2}(p, \kappa + \log 2) = (p/r, \kappa + \log 2 + \log r)$, while $T_{w_1} \circ T_{w_2}(s) = T_{w_1}(p/r, \kappa + \log r) = (p/r, \kappa + \log r + \log 2)$. Since $\log 2 + \log r = \log r + \log 2$ the κ -coordinates agree, but the price adjustment for the rights issue depends on the *pre-split* price in $T_{w_2} \circ T_{w_1}$ and the *post-split* price in $T_{w_1} \circ T_{w_2}$ when the rights issue price depends on the current market price—giving different results. $\square$

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